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APPLICATION CRITERIA
FOR ELECTRICAL EQUIPMENT

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TRANSMISSION ENGINEERING STANDARD

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1.0 PURPOSE AND SCOPE

1.1 The purpose of this chapter is to highlight the practices with respect to the criteria used in the selection and application of substation equipment in the electrical system of Saudi Electricity Company (SEC).

1.2 This standard covers the following substation equipment:

- Power Circuit Breaker
- Metalclad Switchgear
- Power Transformers
- Disconnect Switches
- Power fuses

2.0 POWER CIRCUIT BREAKERS

2.1 Classification of Power Circuit Breakers

2.1.1 Power circuit breakers are classified in terms of their tank construction design as "Dead Tank" and "Live Tank". Dead tank means that the circuit breaker tank and all accessories are maintained at ground potential and the external connections are made through conventional bushings. Live tank means that all metal parts and the interrupting mechanism are mounted inside insulating shell, supported by an insulated column and therefore the tank is live. This column, besides serving as an insulating support, may also act as an access for the operating rod or linkage.

Due to severe service conditions in certain areas of the Kingdom, particularly coastal zones, a creepage distance of 40 mm/kV for outdoor applications has been specified. Most of the conventional live tank circuit breakers cannot meet this requirement in the specified voltage classes. Consequently, the breakers with higher voltage class bushings may have to be applied. Further live tank breakers do not incorporate bushing current transformers like dead tank breakers. Therefore, free standing CTs has to be installed on both sides of the breakers. This results in more exposed porcelain surface and hence requires more maintenance.

For the reasons explained above, the present SEC practice/policy is to accept the Dead Tank circuit breakers for application within 100 km coastal zone A_1 unless otherwise specified. However, live tank circuit breakers meeting the specified requirement of creepage distance can also be used.

2.1.2 In addition, circuit breakers are also classified in terms of interrupting media. Two main types are known as oil and oil-less designs. The oil types comprise "Minimum Oil" and "Bulk Oil" whereas oil-less comprise "Air Blast", "Vacuum" and "SF_6 Gas" circuit breakers.

2.2 Application of Power Circuit Breakers

The oil breakers are no more applied in the electrical system of SEC due to a number of demerits. Air-blast circuit breakers are also not proposed due to the stringent air quality requirements.

Therefore, in this specification, only "SF6 Gas" and "Vacuum" circuit breakers application are described as these are the two common types of breakers normally specified in TMSS and used in SEC network.

2.2.1 Vacuum Circuit Breakers

The application of vacuum circuit breakers in SEC system is limited to 11kV, 13.8 kV, 33 kV and 34.5 kV metalclad switchgear.

Vacuum circuit breakers are generally manufactured for voltages up to 69kV, continuous current ratings up to 4000A, and cover almost all standard interrupting ratings up to 50kA. The pressure in vacuum interrupter is at 10^{-11} bars. The metal-vapor plasma which develops in the contact gap during opening is fully deionized within a few microseconds by diffusion and recombination so that the conduction path very quickly recovers its dielectric strength, and arc quenches at next current zero instant. Vacuum circuit breakers inherently require small contact gap (due to rapid recovery of dielectric strength). The mechanism travel distance is short, therefore resulting in short operating time.

Current chop phenomenon, considered to be a major disadvantage of the vacuum interrupters, has been extensively investigated. It is, however, no longer a concern with state-of-the-art vacuum interrupters using the latest development in the contact material.

Vacuum circuit breakers are particularly suitable for the following applications:

a. For high capacitive current switching (such as unloaded transmission lines and cables) due to the steep rise of the gap dielectric strength in microseconds.

b. For interruption of short-circuit currents following very steep initial rise of Transient Recovery voltage for short-line faults because of reasons highlighted above.

2.2.2 Sulfur Hexafluoride (SF_6) Gas Circuit Breakers

Sulfur Hexafluoride (SF_6) breakers have found extensive applications in SEC electrical system for all voltage levels from 11 kV up to 380 kV and for both indoor and outdoor installations. Basic criteria favoring primarily SF_6 breakers are attributed to SF_6 excellent arc quenching principle, cleanliness and nearly maintenance-free breakers.

a. The SF_6 circuit breakers can cope with all known switching duties and other SEC system ratings and the closed gas system perfectly adapts to environmental requirements. The SF_6 circuit breakers are available for all voltages up to 800 kV, continuous ratings up to 4000 A and symmetrical interrupting rating up to 63 kA.

b. These circuit breakers are generally designed as either puffer (piston) principal or single pressure self-blast technique. Puffer type breakers are employed for high breaking capacities, while the self-blast technique is used for medium-range breaking capacities. Both systems operate fully independently of auxiliary gas supply.

c. SF_6 gas has excellent arc-quenching because of its high thermal conductivity and high dielectric strength. At a temperature of about 1700 °C, SF_6 has a very high specific heat and, accordingly, a high thermal conductivity which promotes cooling of the arc plasma just before and at current zero, and thus facilitates quenching of the arc. The electronegative behavior of SF_6 gas is responsible for the high electric breakdown strength. SF_6 is a very stable compound, inert up to about 500 °C, nonflammable, non toxic, odorless and colorless.

d. SF_6 circuit breakers are available in the following types:

i.	Indoor Design	:	to fit into the metalclad switchgear
ii.	Dead Tank Design	:	for GIS applications
		:	for outdoor substations
iii.	Live Tank Design	:	for outdoor substation layout.
		:	for indoor substation applications both in fixed as well as draw out type design.

3.0 METALCLAD SWITCHGEAR

3.1 General

The metal-clad switchgear is a type of switchgear in which the electrical equipment is located in an enclosing metal structure, generally factory assembled. The equipment required to control an individual circuit, including circuit breaker, busbar, earthing switch, current and potential transformers, controls, measuring instruments and relays, is assembled in separate compartments using metallic partitions. These compartments are designated as low voltage compartment, breaker compartment, busbar compartment and cable compartment.

3.2 Classification of Switchgear

The metalclad switchgear is classified in terms of its use as Indoor or Outdoor.

3.2.1 Indoor Switchgear

Indoor air insulated and SF_6 gas-insulated metalclad switchgear are provided with degree of protection as IP 40/41 which provide protection against solid bodies greater than 1 mm such as fine tools and small wires and vertically falling drops of water as a result of condensation from an AC ducts passing over the switchgear panel. The spaces to be provided in front, back side and extreme ends of the switchgear panels for operation and safe maintenance shall not be less than the spaces as recommended by the manufacturer.

3.2.2 Outdoor Switchgear

Outdoor metalclad switchgear shall be installed in weatherproof housing having degree of protection as IP 54. Outdoor switchgear is provided with either a "standard" or "common" walkway. Standard walkway is used for single lineup whereas common walkway is used for double lineup of the switchgear assembly. Lighting and convenience outlets are to be provided in the walkway. Heaters must also be provided to prevent condensation inside the switchgear panel. To meet the ambient temperature/temperature rise requirements, air conditioning units might sometimes be installed in the switchgear housing. All doors shall be fitted with fire exit hardware.

3.3 Application

Metal-clad switchgear is available up to 40 kV. In SEC network, metalclad switchgear, complying with above mentioned requirements is used for medium voltage systems from 11 kV to 34.5 kV for both indoor and outdoor applications. The type of circuit breakers used is either SF_6 or vacuum. The rated current of the switchgear, at specified local ambient conditions, shall be compatible with the capacity of the power transformer with permissible overloading.

4.0 POWER TRANSFORMERS

4.1 General

4.1.1 Power Transformers shall be either two or multi-winding conventional transformers or auto-transformers. Generally SEC preferred a three-phase unit over the equivalent three single phase units on account of smaller size, lower initial cost and greater efficiency (low operating cost).

All system tie power transformers shall be either three-winding conventional transformers or auto-transformers, in triple ratings having star-star (solidly grounded) connected main windings and a delta connected tertiary winding unless otherwise specified. The choice between a three-winding transformer and an auto-transformer is purely a matter of economics. The smaller the transformation voltage difference the smaller will be the initial cost of the autotransformer.

4.2 Ratings

Refer to Engineering Standard TES-P.119.02 entitled "Basic Design Aspects" which indicates the ratings of the transformers, utilized by SEC Operating Areas.

4.3 Auto-Transformers

The auto transformers are generally used for transforming from one transmission voltage to another when the ratio is 3:1 or less. These transformers are normally connected in Y with the neutral solidly grounded and provided with a closed delta tertiary winding. For details tertiary winding, refer to Section 4.5.

The standard voltage ratings of auto-transformers in common use in SEC system are, 380/230 kV, 380/132 kV, 380/110 kV, 380/115 kV and 230/115 kV. The wye connected auto-transformer provides an "in-phase" angular relationship between primary and secondary voltages. Advantages of auto-transformers, when compared with separate-winding transformers include lower losses, better voltage regulation, smaller size, lighter weight and lower cost. The main disadvantages are lower impedance and complex design problems in providing higher range of tapping. Due to their low impedance, auto-transformers are likely to fail under high short circuit currents unless adequately protected.

4.4 Three-Winding Transformers

In SEC System there are two types of conventional three-winding transformers. One type is with HV, LV & TV windings connected in star, star and delta. The HV neutral is solidly earthed while the secondary neutral is solidly grounded or through earthing resistor. The Delta connected stabilizing winding may also be designed to supply an external load. Each of the three windings is interlinked with the magnetic leakage flux of the other windings. Load currents in one circuit affect voltages in other windings. The arrangement of the windings in a multi-winding transformer is, therefore, made in such a way as to obtain optimum value of voltage regulation. The other type of three-winding transformer used in SEC system is with HV, LV1 & LV2 windings.

4.5 Tertiary Winding

4.5.1 The application and uses of a delta connected tertiary winding are as follows:

a. Act as a stabilizing winding and provide a path for zero sequence currents. Thus the third harmonic voltages are eliminated by the circulation of third harmonic currents in the short circuit path of the delta windings.

b. In substation applications, to supply auxiliary loads.

4.5.2 The tertiary winding shall be designed to be strong enough mechanically to have the requisite thermal capacity and to have sufficient impedance with

respect to the two main windings to be able to withstand the electromagnetic effects of short circuit across main windings.

Industry Standards specify tertiary winding rating to be normally one third of that of the main winding. In some cases in SEC network, the tertiary winding is used to supply station service loads. The maximum loading is limited to 2 MVA. A satisfactory value of the rating of the tertiary winding will be dependent upon the impedances between windings and the magnitudes of the short-circuit currents that will flow in the tertiary winding.

4.5.3 The tertiary winding leads are normally brought out through three fully rated bushings and grounded at one corner via a removable strip, if not used for supplying station service loads unless otherwise specified.

4.6 Rating Criteria

The specified forced ratings correspond to single stage or two-stage forced air cooling methods. Each stage of forced cooling shall enhance the MVA rating by 33% of ONAN rating. Accordingly, the triple rating of the transformer shall be stated as 60%/80%/100% for ONAN/ONAF1/ONAF2 or OFAF respectively, unless otherwise specified.

4.7 Types of Windings

4.7.1 The principal three-phase connections of transformer windings are star (Y-connected) and delta (D-connected). The winding connections chosen influence the design of the transformer and the amount of material required.

4.7.2 Advantages

a. Y-connected Winding

i. It is more economical for a high voltage winding.
ii. It has a neutral point available.
iii. It permits direct earthing or earthing through impedance.
vi. It permits reduced insulation level (graded insulation).
v. It permits the winding taps to be located at the neutral end of each phase.
vi. It permits single phase loading.

b. Delta-connected Winding

i. It is more economical for a high-current, low voltage winding.
ii. In combination with a star-connected winding, it reduces the zero-sequence impedance in that winding.

4.8 Vector Group (Phasor Symbols)

4.8.1 Vector group for two-winding transformer shall be Dyn1 or Dyn11 / YNd1 corresponding to a phase displacement of -30º or +30º respectively.

The advantage of delta/star winding is that third harmonic voltages are eliminated by the circulation of third harmonic current in the primary delta. However, the delta winding may be mechanically weak in the case of a step down transformer with a very high primary voltage, or in case of a transformer of small rated output. In SEC network, these transformers are best suited for application on system voltages of 132kV, 115 kV, 110 kV, 69 kV & 33 kV with ratings of 20 up to 67 MVA. Transformers with delta/star connections are equally suited for step up operations as both windings are utilized under best conditions.

4.8.2 The vector group for three-winding conventional transformers shall be YNyn0d1. Also, owing to the relatively large conductors, the electrostatic capacitance between turns is high so that the severity of the stress due to transient voltage waves striking the windings is lessened. The tertiary windings provide the necessary magnetizing ampere turns for counteracting the effects of third harmonic voltages and currents.

4.8.3 For auto-transformers the vector group shall be YNa0d1.

4.9 Parallel Operation

Transformers are in parallel operation if they are connected in parallel on at least two sides.

4.9.1 For the satisfactory parallel operation of transformers, the following conditions must be satisfied in order to avoid dangerous transient currents:

a. The same vector group and phase angle number.
b. The same voltage ratios within the limits prescribed in clause 4.9.2 below.
c. The percentage impedance shall be approximately the same with permissible discrepancies within the limits prescribed in clause 4.9.3 below.
d. The same polarity
e. The same phase sequence
f. The same tap changer size and tapping range
g. The same power rating, preferably.

4.9.2 It is very desirable that the voltage ratios of any two or more transformers operating in parallel must be the same. If there is any difference, a circulating current will flow in the secondary windings of the transformers before they are connected to any external load. The ratio error in transformers for parallel operation shall not exceed ± 0.5%. In modern

transformers a circulating current of the order of 5% of the full load current is generally accepted without any fear of serious overheating.

4.9.3 In case of unequal impedances, the load will be divided in inverse proportion to the magnitude of the impedance. From this stand point, it has been generally accepted that transformers can be paralleled successfully if the name plate impedance of one does not differ by more than 7.5% from the name plate impedance of the other at principal tapping.

4.10 Cooling

In power transformers, cooling is essential in order not to exceed the temperature rise of oil and winding at maximum ambient temperature as described in 01-TMSS-01 and to prevent the accelerated loss of life of insulation due to excessive internal temperature.

The most common types of cooling arrangements are specified in industry standards and identified by a system of symbols, which indicate the cooling medium in contact with the winding, the cooling medium in contact with the external cooling system and the kind of circulation for each. The symbols for oil, air and water are 'O', 'A' and 'W' and 'N' and 'F' for 'natural' and 'forced' are the kind of circulation respectively. Based upon the above analogy of systems, the various types of cooling method are 'ONAN', 'ONAF', 'OFAN', 'OFAF', 'ONWF', 'OFWF', etc.

The forced air cooling system reduces the size of the radiating surfaces, while the forced oil system enables the windings and core to be economically designed. In short, forced cooling methods are effective in reducing the size and cost of the larger transformers. It is normally possible to obtain 67% more capacity above ONAN rating by employing forced air and forced oil cooling methods. Both methods can be automatically controlled in one or two stages from either top oil temperature indicator or winding temperature indicator or both.

SEC, however, specifies the following methods of cooling:

ONAN : Natural oil circulation, natural air flow.
ONAF : Natural oil circulation, forced air flow (by using fans) in single or two stages that is ONAF-1 & ONAF-2.
OFAF : Forced oil circulation (by means of pumps), forced air flow (by means of fans).

For triple rated power transformers the third rating (corresponding to second stage of forced cooling) may be based upon either forced air cooling or forced oil cooling.

In order to ensure the operational reliability of the forced cooling system, SEC specifies 25% redundancy in forced air cooling capacity and 100% redundancy in pumps for forced oil cooling methods.

4.11 Tap Changer

4.11.1 On load tap changer shall be provided on HV windings. The number of taps, number of ± steps and step voltage shall be as per SEC specified requirements in a particular project. The control of tap changer shall be automatic unless otherwise specified and with supervisory control.

4.11.2 For inter-tie power transformers, operation of tap changers shall be operated manually by Power Control Center (PCC) to allow control of VAR flow.

The voltage regulation of the transformer shall be within the specified tapping range of the OLTC at principal tap. Calculations of voltage regulation, based on guaranteed value of percentage impedance at rated load (rated MVA), 0.8 power factor, and principal tapping of the OLTC, shall be submitted along with the test report for SEC review and approval. In addition, the transfer surge voltage calculations for tertiary winding of transformer (if applicable) shall be furnished for SEC review.

4.12 Surge Arresters

HV and LV terminals of all transformers (69 kV and above) shall be protected by surge arresters, except in cases where the transformer bushings are housed in cable termination boxes or when the transformer bushings (SF6/Oil) terminate directly onto a Gas Insulated Switchgear (GIS).

4.13 Graded Insulation

When a star connected transformer is connected to an effectively grounded system, some cost savings can be realized by grading the insulation to neutral and using a reduced BIL neutral bushing. However, for 11kV, 13.8kV, 33kV and 34.5kV star windings are insulated uniformly as the cost savings of graded insulation is minimal. Also for the advantage of bushing interchangeability, the neutral shall be fully insulated.

4.14 Treatment of Mineral Oil before filling

Before filling the oil into any electrical equipment concerned, the unused mineral insulating oil as per 54-TMSS-01 shall be treated and after treatment the oil shall be tested by the test methods described hereunder and shall meet the following specific requirements as shown in Table 12.1 below.

Table 12-1

Characteristics	Test Method		Permissible Value	
	IEC	ASTM	IEC	ASTM
Dielectric Strength	IEC 60156	D 1816	70 kV @ 2.5 mm	28 kV @ 1.02 mm 56 kV @ 2.03 mm
Neutralization Number	IEC 62021-1	D 974	0.01 mg KOH/g (max)	0.03 mg KOH/g (max)
Water Content	IEC 60814	D 1533	10 ppm (max)	
Interfacial Tension	AFNOR NF ISO 6295	D 971	40 dynes/cm (min)	

5.0 DISCONNECT SWITCHES & GROUNDING SWITCHES

Disconnect switches are used primarily for isolating the equipment such as live apparatus and buses for testing and/or maintenance purposes. These are also used for sectionalizing the electric circuits. The disconnect switches are designed to carry the rated load currents and remain closed during flow of fault current. Generally these devices are not designed to break the load currents. Fault currents in excess of specified ratings may cause the disconnects to blow open by the magnetic forces of the short circuit current. High pressure contacts are used to prevent this opening. SEC specifies silver plated contacts. High contact pressures ensure good self-cleaning action - a characteristic of paramount importance for SEC application on account of severe atmospheric pollution in certain areas. Effective measures have to be taken for contact protection against sulfur contamination when the switches are installed in the vicinity of the gas fields or the chemical industry. The performance of disconnect depends not only on the magnitude of current but also on the speed of operation, wind direction and clearance to other energized or grounded components. Disconnects are available in voltage ranges from 2.4 kV to 765 kV and current ratings up to 4000 A.

SEC applies three-pole gang-operated disconnect switches. These can be manually actuated or in remotely operated installations, motor or compressed-air drives. The outdoor line and the 380kV disconnect switches preferred by SEC is motor operated. The disconnect switches are single-column pantograph or semi-pantograph, two-column single-break with horizontal or vertical movement or a three-column double-break with horizontal movement (blade rotating about central column) for both indoor and outdoor applications. The disconnect switches are mounted in horizontal, vertical or inverted configurations.

In certain applications of SEC, disconnect switches are required with load breaking capability to interrupt limited values of certain currents. These interruptions shall be performed without any ferro-resonance problems. The applications include interruption of the required value of capacitive currents in 380 kV GIS, interruptions of transformer magnetizing inrush currents and interruption of live bus transfer currents in double bus bar configurations. Depending upon the magnitude of the currents, the disconnect switches may or may not be equipped with specific auxiliary devices for successful interruption.

The grounding switches for GIS applications shall be essentially motor operated to ensure high speed operation. In the metalclad switchgear the grounding switches are incorporated in the structure and are manually operated, and interlocked with the circuit breakers.

6.0 POWER FUSES

6.1 General

A fuse protects the electric circuit and the equipment connected to it, from over-current damage within the limits of its ratings (time-current characteristics). The performance of the fuse depends upon the correctness of the application in addition to manufacturing accuracy and post-installation care.

6.2 Types of Power Fuses

High Voltage (above 1000V) Power Fuses are available in two basic types :

- Expulsion Power Fuses
- Current Limiting Power Fuses

6.2.1 Expulsion Power Fuses

Expulsion fuses are the fuses in which the arc is extinguished by the expulsion effects of the gases produced by the arc. When the fuse link melts the ensuing arc impinges on the wall of the fibrous tube holder generating gas which provides the expulsion action needed to extinguish the fault current.

These fuses have no energy-limiting ability and require a natural current-zero crossing to successfully interrupt a short circuit current. Consequently a large amount of energy is fed into the fault until the instant of current zero. These types of fuses are used as external fuses for the capacitor bank protection in the SEC network.

Expulsion fuses are either "Un-vented" or "Vented".

a. Un-vented Expulsion Fuses: These fuses expel hot gases, flames and solid particles with a very loud noise. Such types of fuses are to be used at remote substations away from human habitation.

b. Vented Expulsion Fuses: These Boric acid or solid material fuses produce arc products in the form of noncombustible steam which is condensed and cool the arc of the melting fuse link. Vented fuses are superior to their un-vented counterparts in terms of maximum continuous current rating and symmetrical interrupting rating.

6.2.2 Current Limiting Power Fuses

This type of fuse generally is constructed of silver wire or ribbon fusible elements, often in parallel, spirally wound on a core or spider and packed in quartz-sand filler in a sealed cylindrical glass or epoxy-glass container. Provisions for suitable electrical connection are provided at the ends. When the operation takes place under the high-fault conditions**,** the fusible element melts almost instantaneously at a series of reduced sections all along the length. The resulting arc dissipates its heat rapidly into the surrounding sand.

These fuses are used for the protection of station service transformers in conjunction with RMU and capacitor banks at remote system based on case-by-case studies.

The current limiting fuses are characterized by:

a. High current interrupting ability.

b. Operation is noiseless and there is no expulsion of the arc or arc products. Thus the fuse can be packaged into relatively confined space.

c. In the current limiting mode of operation, the interruption time is very fast, one-half cycle or less.

d. Current-limiting action and fast operation reduce the amount of fault energy (I^2t) let through in to the failed equipment, thereby reducing the resultant damage.

e. Current Limiting fuses have three distinct classes:

i. General purpose: These fuses are designed to clear fault current over a broad range. They are defined as a fuse capable of interrupting all currents from the maximum interrupting current down to current causing melting of the element in one hour. Current-limiting fuses inherently are excellent fault-current interrupters in the high current range. Typical general purpose fuses operate in the current limiting mode at fault currents equal to approximately twenty-five times the rated current or larger.

ii. Back-up: These fuses as defined are capable of interrupting all currents from the rated maximum interrupting current down to the minimum interrupting current as given by the

manufacturer. The low current clearing must be accomplished by another auxiliary device, most commonly an expulsion fuse. In this case, the time-current characteristics are a composite of the two fuses.

iii. Full Range : These current limiting fuses capable of breaking under specified conditions and behavior, all currents that cause melting of fuse element(s) up to its rated maximum breaking current.

6.3 Service Conditions

Power fuses shall be suitable for operations at the specified ambient conditions per TES-P-119.02. Manufacturer's published fuse characteristics are usually based on an ambient temperature of 25 °C or 40 °C and no initial loading. Any increase in ambient temperature at any load current will reduce the melting time for a given current. In such cases, manufacturers should be consulted for applying the relevant correction factors. Typical correction factors for dielectric strength and rated continuous current, under special service conditions are available in ANSI C37.40 and IEC 60282 standards.

6.4. Selection of Rated Voltage & Current

6.4.1 Rated Voltage

The voltage rating of fuse shall be equal to or exceed the highest system voltage for the equipment. The rated voltage of a fuse shall be selected from the standard values given in Section 2 of ANSI C37.46, or Table 3 of IEC 60282-1 and 2.

6.4.2 Selection of Rated Current

The standard values of the rated currents can be selected from Section 2.0 of ANSI C37.46 or IEC specified R10 series.

6.5 Clearances

For multi-pole arrangement of fuses, when the distance between the poles is not fixed by the construction, the poles should be mounted with clearances not less than those specified by the manufacturer.

6.6 General Application Requirements

The rated current of the fuse-link should be selected with due regard to the following parameters:

6.6.1 Normal and permissible overload currents of the circuit, including sustained harmonics.

6.6.2 Transient phenomena in the circuit related to switching such equipment as transformers, motors, or capacitors.

6.6.3 Co-ordination with other protective devices, if any.

6.6.4 Ambient and mounting conditions

The rated current of a fuse-link is usually higher than the normal service current. If the rated current of the fuse-link is less than that of the fuse-base or fuse-carrier, the effective current rating of the fuse is that of the fuse-link. Recommendations for selection are usually provided by the manufacturer.

6.7 Power Fuse Application for Transformers

6.7.1 General

The over-current device connected on the line (primary) side of a transformer provides short-circuit protection whereas the over-current device connected on the load (secondary) side of a transformer provides overload protection for the transformer. In the selection of the current rating of a high voltage fuse the following objectives shall be taken into consideration:

a. Permit loading of the transformer per over loading criteria of user (SEC)

b. Withstand combined magnitude of magnetizing inrush and load pick up currents after short time (up to 1 minute) interruption at the substation. Please refer to clause 6.7.3, item a.

c. Properly coordinate with all upstream and downstream over-current protecting devices.

d. Protect the substation bus from faults at/or within the transformer.

e. Protect the transformer from through faults.

f. Provide earliest detection and clearing of internal transformer faults.

6.7.2 Coordination

It must be ensured that the selected fuse for a particular application in an interconnected system must operate correctly relative to the other fuses or over-current devices installed in the system. For transformer application, the time/current characteristics of high voltage fuse should have:

a. Relatively high operating current in the 0.1 s region so as to withstand transformer inrush current and give good co-ordination with protection devices in the secondary side.

b. Relatively low operating current in the 10 s region so as to ensure rapid clearance of transformer winding faults, secondary side faults and, if applicable, primary side earth faults, and to give good co-ordination with over-current protective devices on the source side.

6.7.3 A typical transformer application involving current limiting high voltage fuse links and possible protective devices on both source and load sides, as per IEC-60787, is illustrated in Fig. 12-01. The transformer shall be considered for its particular duty with reference to the full load current, permissible overload current and inrush current. Following criteria shall be considered for adequate fuse coordination.

a. The transformer magnetization inrush current, both in magnitude and duration, shall be less than the minimum (melting) time-current characteristics of the fuse. This means that the minimum pre-arcing time/current characteristics of the primary fuse shall be to the right of the point 'A'. As a conservative approach, the inrush current shall be taken as 10-12 times the highest continuous current of the transformer for duration of 0.1 second.

b. The current rating of the HV fuse link shall exceed the full load current of the transformer by an amount to account for the following:

i. permissible overloading under service conditions
ii. operations beyond the specified service conditions
iii. fuses mounted in enclosures, with reference to specified temperature-rise limits for fuse-links.

The minimum ratio between the fuse rating and the transformer rated current is usually taken as 1.5: 1.

c. The minimum time-current characteristics of the fuse shall be such as to ensure relatively low operating current in the 10 seconds region to ensure rapid clearance of transformer winding faults, secondary side faults, and primary side earth faults (if applicable), and give good coordination with over-current protective device(s) on the source side. To accomplish this, operating current in the 10 seconds region, according to IEC 60787, shall be equal to or less than 6 times the normal operating current of the fuse.

d. For overall coordination between primary side fuse and secondary side protective device(s), the intersection point 'B' shall occur at a value higher than that of the maximum fault current on the load side.

6.7.4 As a general rule the minimum time-current characteristics of a fuse (or melting I^2t, if applicable) shall be higher than the total clearing time-current characteristics (or clearing I^2t) of any over-current device(s) on the secondary side (load side). Similarly the total clearing curve of the primary fuse shall be kept below the minimum time-current curve of any over-current device(s) on

the source side (or clearing I^2t of the primary fuse shall be less than the melting I^2t of any source side protective device(s).

6.7.5 In case of parallel transformers, the use of individual primary fuses shall be avoided.

6.7.6 While selecting fuse for power transformer, the manufacturer's heating curve shall also be consulted. The total clearing time-current characteristics of the selected fuse shall be below the transformer heating curve for all currents up to the maximum fault current through the transformer.

6.8 Metalclad Switchgear, 11kV, 13.8 kV, 33 kV and 34.5 kV

Power fuses of current limiting type are used on the primary side of the potential transformers of medium-voltage switchgear.

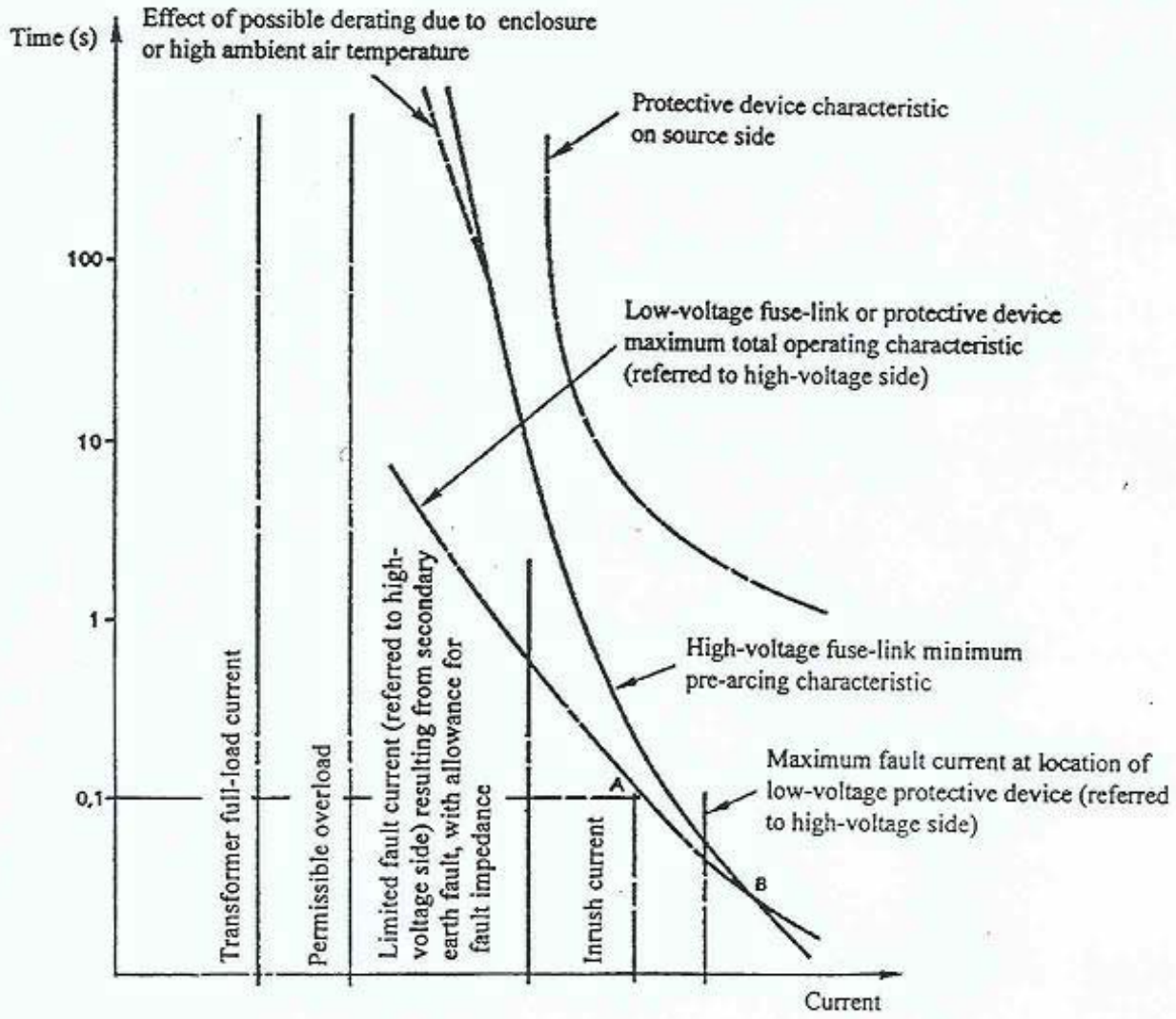

Figure 12-01: Characteristics Relating to Protection of HV/LV Transformer Circuit.

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